

DIKSHIT RAJ PABATHI

Azure Snowflake Data Engineer

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PROFILE SUMMARY

- Overall 10+ years of professional IT experience and over 5+ years in Data Engineering, and 4+ years in Data warehouse.
- Proficient in designing and executing scalable data ingestion pipelines leveraging Azure Data Factory, PySpark, Sqoop & StreamSets.
- Experience in Data pipeline creation using Delta Lake, Delta Tables, Delta Live Tables, Data Catalogues, and Delta Lake API.
- Proficiency in real-time data streaming pipeline creation utilizing Azure Event Hub and Databricks Auto Loader.
- Demonstrated skills in data pipeline development with PySpark, Databricks, and Databricks job API.
- Experience in Data pipeline implementation using Pandas DataFrame, Spark DataFrame, and RDDs.
- Solid expertise in Spark performance tuning, SparkSQL, Spark Streaming in Bigdata & Azure Databricks.
- Strong proficiency in Spark job optimization and utilization of Azure Synapse Analytics for large-scale data processing and analytics.
- Developed optimized Data Warehouse solutions employing Azure Synapse.
- Implemented server-less solutions leveraging Azure Functions.
- Experience in workflow creation using Azure Logic Apps.
- Proficient in Python and Scala within the Spark framework.
- Experienced in utilizing file formats like ORC, Parquet, Avro, and delimited.
- Implemented Hive performance tuning techniques and User-Defined Functions (UDFs).
- Experience with real-time data injection techniques using Apache Kafka.
- Hands-on experience in implementing data pipeline solutions using an array of technologies, including Hadoop, Azure, ADF, Synapse, PySpark, Map-Reduce, Hive, Tez, Python, Scala, Azure functions, Logic Apps, StreamSets, ADLS Gen2, and Snowflake.
- Proven skills in data pipeline development with Snow SQL, Snowflake Integrated services, and Snow Pipe.
- Utilized Snowflake Clone and Time Travel features.
- Active participation in the enhancement and maintenance of Snowflake database applications.
- Built Logical and Physical data models for Snowflake based on necessary changes.
- Experience in leveraging Snowflake Multi-Cluster Warehouses.
- Proficiency in designing cloud-based data warehouse solutions using Snowflake and optimizing schemas, tables, and views for effective data storage and retrieval.
- Ensured data quality and integrity through validation, cleansing, and transformation operations.
- Deep understanding of Snowflake Database, Schema, and Table structures.
- Implemented data encryption using Voltage and Azure Key Vaults.
- Experience in developing large-scale data pipelines using Spark and Hive.
- Used Apache Sqoop for data import and export from HDFS.
- Scheduled Hadoop jobs with Apache Oozie and imported/exported data using SQOOP from HDFS to Relational Database Systems.
- Improved query performance in Hive using bucketing and partitioning techniques and have extensive hands-on experience tuning Spark Jobs.
- Collaborative approach to working with data analysts and stakeholders to implement appropriate data models and structures.
- Proficient in Agile methodologies and project management tools like JIRA/ADO.
- Implemented CI/CD data pipelines and worked with DevOps teams for automated pipeline deployment.
- Developed Machine Learning data pipelines with Azure ML Flow.
- Implemented production data pipelines using Apache Airflow, YAML, and Terraform scripts.
- Experience in using version control tools such as GitHub, Azure Dev Ops, Bitbucket, GitLab, and ARM templates.
- Developed data visualizations with Power BI & Power BI DAX.

- Demonstrated skills in shell scripting.
- Built Data Warehouse solutions using Snowflake & Star Schema.
- Experience in data pipeline creation with Informatica & PL SQL.
- Implemented SQL Analytical Functions & Window Functions.

TECHNICAL SKILLS

Azure Services	Azure data Factory, Azure Data Bricks, snowflake, Logic Apps, Functional App, Snowflake, Azure DevOps
Big Data Technologies	MapReduce, Hive, Teg, Python, PySpark, Scala, Kafka, Spark streaming, Oozie, Sqoop, Zookeeper
Hadoop Distribution	Cloudera, Horton Works
Languages	SQL, PL/SQL, Python, HiveQL, Scala.
Web Technologies	HTML/HTML5, XML, CSS/CSS3, Bootstrap, JSP, Ajax, NodeJS, Express, ReactJS, AngularJS, Angular 2.0 and above
Operating Systems	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.
Build Automation tools	Ant, Maven
Version Control	GIT, GitHub.
IDE & Build Tools, Design	Eclipse, Visual Studio.
Databases	MS SQL Server 2016/2014/2012, Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Cosmos DB

EDUCATION

- Bachelor's in Computer Science Engineering from Jawaharlal Nehru Technological University, India.

PROFFESIONAL EXPERIENCE

PNC, New York metropolitan Area

Azure Snowflake data engineer | Jun 2022 – Till Now

Responsibilities:

- Designed and deployed scalable data ingestion pipelines with Azure Data Factory, integrating data from a range of sources like SQL databases, CSV files, and REST APIs.
- Constructed data processing workflows using Azure Databricks, utilizing Spark for distributed data processing and transformation tasks.
- Assured data quality and integrity by performing data validation, cleansing, and transformation operations through Azure Data Factory and Databricks.
- Designed and deployed a cloud-based data warehouse solution using Snowflake on Azure, capitalizing on its scalability and performance capabilities.
- Established and optimized Snowflake schemas, tables, and views to support efficient data storage and retrieval for analytics and reporting needs.
- Collaborated with data analysts and business stakeholders to understand requirements and established suitable data models and structures in Snowflake.
- Developed and optimized Spark jobs for data transformations, aggregations, and machine learning tasks on extensive data sets.
- Utilized Azure Synapse Analytics to incorporate big data processing and analytics capabilities, facilitating seamless data exploration and insights generation.
- Configured event-based triggers and scheduling mechanisms for automating data pipelines and workflows.
- Implemented data lineage and metadata management solutions to monitor data flow and transformations.
- Identified and addressed performance bottlenecks in data processing and storage layers, enhancing query execution and reducing data latency.

- Employed partitioning, indexing, and caching strategies in Snowflake and Azure services to improve query performance and cut down processing time.
- Performed performance tuning and capacity planning to ensure scalability and efficiency of the data infrastructure.
- Developed a CI/CD framework for data pipelines using Jenkins.
- Collaborated with DevOps engineers to develop automated CI/CD and test-driven development pipelines using Azure as per client requirements.
- Applied hands-on programming expertise in scripting languages like Python and Scala.
- Participated in running all Hive scripts through Hive on Spark and some through SparkSQL.
- Collaborated on ETL tasks, maintained data integrity, and verified pipeline stability.
- Gained hands-on experience in using Kafka, Spark streaming, to process streaming data in specific use cases.
- Developed a data pipeline using Kafka, Spark, and Hive for data ingestion, transformation, and analysis.
- Designed and implemented real-time data processing solutions using Kafka and Spark Streaming, enabling the ingestion, transformation, and analysis of high-volume streaming data.
- Developed Spark core and Spark SQL scripts using Scala for expedited data processing.
- Worked with JIRA for project reporting, and creating subtasks for Development, QA, and Partner validation.
- Experienced in the full range of Agile ceremonies, from daily stand-ups to internationally coordinated PI Planning.

Environment: Azure Databricks, Data Factory, Snowflake, Logic Apps, Functional App, Snowflake, MS SQL, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, shell scripting, GIT, JIRA, Jenkins, kafka, ADF Pipeline, Power BI.

Client: Kroger, Columbus, Ohio

Role: Azure Snowflake data engineer | Apr 2021 – May 2022

Responsibilities:

- Implemented end-to-end data pipelines with Azure Data Factory, executing ETL operations from various sources into Snowflake.
- Crafted and executed data processing workflows using Azure Databricks, utilizing Spark for large-scale data transformations.
- Constructed scalable and optimized Snowflake schemas, tables, and views to support complex analytics queries and reporting requirements.
- Developed data ingestion pipelines using Azure Event Hubs and Azure Functions for real-time data streaming into Snowflake.
- Employed Azure Data Lake Storage as a data lake for storing raw and processed data, implementing data partitioning and data retention strategies.
- Utilized Azure Blob Storage for efficient storage and retrieval of data files, implementing compression and encryption techniques to optimize storage costs and data security.
- Integrated Azure Data Factory with Azure Logic Apps for orchestrating complex data workflows and triggering actions based on specific events.
- Implemented data governance practices and data quality checks using Azure Data Factory and Snowflake, ensuring data accuracy and consistency.
- Executed data replication and synchronization strategies between Snowflake and other data platforms using Azure Data Factory and Change Data Capture techniques.
- Developed and deployed Azure Functions for data preprocessing, data enrichment, and data validation tasks in data pipelines.
- Implemented advanced analytics and machine learning workflows using Azure Machine Learning and Snowflake, enabling predictive analytics and data-driven insights.
- Designed and implemented data archiving and data retention strategies using Azure Blob Storage and Snowflake's Time Travel feature.
- Developed custom monitoring and alerting solutions using Azure Monitor and Snowflake Query Performance Monitoring (QPM) for proactive identification and resolution of performance issues.
- Integrated Snowflake with Power BI and Azure Analysis Services for creating interactive dashboards and reports, enabling self-service analytics for business users.

- Optimized data pipelines and Spark jobs in Azure Databricks for improved performance, including tuning of Spark configurations, caching, and leveraging data partitioning techniques.
- Implemented data cataloging and data lineage solutions using tools like Azure Purview and Apache Atlas to provide a comprehensive understanding of data assets and their relationships.
- Collaborated with cross-functional teams including data scientists, data analysts, and business stakeholders to understand data requirements and deliver scalable and reliable data solutions.

Environment: Azure Databricks, Data Factory, Logic Apps, Snowflake, Functional App, Snowflake, MS SQL, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, shell scripting, GIT, JIRA, Jenkins, kafka, ADF Pipeline, Power Bi.

Ameritas

Big data Developer | Apr 2020 – Mar 2021

Responsibilities:

- Developed and maintained data pipelines using Sqoop, Flume, and Kafka to ingest, transform, and process customer behavioral data for analysis.
- Performed data aggregation and analysis on large-scale datasets using Apache Spark, Scala, and Hive, resulting in improved insights for the business.
- Utilized big data ecosystems such as Hadoop, Spark, and Cloudera to load and transform large sets of structured, semi-structured, and unstructured data.
- Integrated HBase with Hive on the Analytics Zone, creating and optimizing HBase tables for faster and more efficient querying of data.
- Utilized Hive queries and Spark SQL to analyze and process data, meeting specific business requirements and simulating MapReduce functionalities.
- Implemented automation for deployments using YAML scripts, resulting in faster and more efficient builds and releases.
- Migrated data from RDBMS (Oracle) to Hadoop using Sqoop for processing, enhancing data management and processing capabilities.
- Utilized JIRA to manage issues and project workflow, improving project organization and efficiency.
- Collaborated with team members to identify and resolve JVM-related issues, resulting in better performance and system stability.
- Utilized Git as a version control tool to maintain the code repository, ensuring better code management and tracking of changes.

Environment: Sqoop, MYSQL, HDFS, Apache Spark Scala, Hive Hadoop, Cloudera, HBASE, Kafka, MapReduce, Zookeeper, Oozie, Data Pipelines, RDBMS, EC2, Python, PySpark, shell script, Ambari, JIRA.

Charter Communications, Englewood, CO

Big data Developer | May 2019 – Mar 2020

Responsibilities:

- Prepared an ETL framework using Sqoop, Pig, and Hive to bring in data from various sources and make it available for consumption.
- Processed HDFS data and created external tables using Hive, along with developing scripts for table ingestion and repair for reuse across the project.
- Developed ETL jobs using Spark and Scala to migrate data from Oracle to new MySQL tables.
- Utilized Spark (RDDs, DataFrames, Spark SQL) and Spark-Cassandra Connector APIs for various tasks, including data migration and business report generation.
- Developed a Spark Streaming application for real-time sales analytics.
- Analyzed source data, efficiently handled data type modifications, and used Excel sheets, flat files, and CSV files to generate Power BI ad-hoc reports.
- Analyzed SQL scripts and designed solutions using PySpark.

- Extracted data from various data sources into HDFS using Sqoop.
- Handled data import from various sources, performed transformations using Hive and MapReduce, and loaded data into HDFS.
- Extracted data from MySQL into HDFS using Sqoop.
- Implemented automation for deployments using YAML scripts for streamlined builds and releases.
- Worked with Apache Hive, Apache Pig, HBase, Apache Spark, Zookeeper, Flume, Kafka, and Sqoop.
- Implemented data classification algorithms using MapReduce design patterns.
- Extensively worked on creating combiners, partitioning, and distributed cache to enhance the performance of MapReduce jobs.
- Utilized Git and GitHub repositories to maintain the source code and enable version control.

Environment: Hadoop, Hive, spark, PySpark, Sqoop, Spark SQL, Shellsript, Cassandra, YAML, ETL.

Bank of America, Plano, TX

Big data Developer | May 2018 – Apr 2019

Responsibilities:

- Experience in developing complex store procedures, efficient triggers, required functions, creating indexes and indexed views for performance.
- Excellent Experience in monitoring SQL Server Performance tuning in SQL Server
- Expert in designing ETL data flows using SSIS, creating mappings/workflows to extract data from SQL Server and Data Migration and Transformation from Access/Excel Sheets using SQL Server SSIS.
- Efficient in Dimensional Data Modeling for Data Mart design, identifying Facts and Dimensions, and developing, fact tables, dimension tables, using Slowly Changing Dimensions (SCD).
- Experience in Error and Event Handling: Precedence Constraints, Break Points, Check Points, Logging.
- Experienced in Building Cubes and Dimensions with different Architectures and Data Sources for Business Intelligence and writing MDX Scripting.
- Thorough knowledge of Features, Structure, Attributes, Hierarchies, Star and Snow Flake Schemas of Data Marts.
- Good working knowledge on Developing SSAS Cubes, Aggregation, KPIs, Measures, Partitioning Cube, Data Mining Models and Deploying and Processing SSAS objects.
- Experience in creating Ad hoc reports and reports with complex formulas and to query the database for Business Intelligence.
- Expertise in developing Parameterized, Chart, Graph, Linked, Dashboard, Scorecards, Report on SSAS Cube using Drill-down, Drill-through and Cascading reports using SSRS.
- Flexible, enthusiastic and project oriented team player with excellent written, verbal communication and leadership skills to develop creative solutions for challenging client needs.

Environment: MS SQL Server 2016, Visual Studio, SSIS, Share point, MS Access, Team Foundation server, Git.

Zillow Group, Seattle, WA

Data warehouse Developer | May 2016 – Apr 2018

Responsibilities:

- Created Stored Procedures, Triggers, Indexes, User defined Functions, Constraints etc on various database objects to obtain the required results.
- Import & Export of data from one server to other servers using tools like Data Transformation Services (DTS)
- Wrote T-SQL statements for retrieval of data and involved in performance tuning of TSQL queries.
- Transferred data from various data sources/business systems including MS Excel, MS Access, Flat Files etc to SQL Server using SSIS/DTS using various features like data conversion etc. Also Created derived columns from the present columns for the given requirements.

- Created logging for ETL load at package level and task level to log number of records processed by each package and each task in a package using SSIS.
- Developed, monitored and deployed SSIS packages.
- Supported team in resolving SQL Reporting services and T-SQL related issues and Proficiency in creating different types of reports such as Cross-Tab, Conditional, Drill-down, Top N, Summary, Form, OLAP and Sub reports, and formatting them.
- Creation, manipulation and supporting the SQL Server databases.
- Involved in the Data modeling, Physical and Logical Design of Database
- Helped in integration of the front end with the SQL Server backend.
- Provided via the phone, application support. Developed and tested Windows command files and SQL Server queries for Production database monitoring in 24/7 support.

Environment: IBM WebSphere DataStage EE/7.0/6.0 (Manager, Designer, Director, Administrator), Ascential Profile Stage 6.0, Ascential QualityStage 6.0, Erwin, TOAD, Autosys, Oracle 9i, PL/SQL, SQL, UNIX Shell Scripts, Sun Solaris, Windows 2000.

JP MORGAN CHASE, West Haven, CT

Data warehouse Developer | March 2014 – Apr 2016

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